Task 10:

Develop an express web application that can interact with REST API to perform CRUD operations on student data. (Use Postman)

Implementation of Steps:

1. Set up the server:

First, initialize your project (create an folder e.g.- task10) and run in terminal
 npm init -y
npm install express
npm install mysql2

Step 2:

- Create a file : filename.js
- Then implement CRUD operations.

3. Install postman application

- download the official installer from the Postman website for Windows 64-bit or ARM 64-bit, run the .exe file, and follow the prompts.

- The installation automatically sets up the desktop app, which supports API testing and development, allowing you to sign in or use it immediately.

Program :

app.js

```
const express = require('express');
const mysql = require('mysql2');
const app = express();
app.use(express.json());

// Create the connection pool
const pool = mysql.createPool({
    host: '127.0.0.1',
    user: 'root',
    password: ' ',   // your mysql pwd
    database: ' ', // your db name
    port: 3307   // your mysql port no
}).promise(); // Use .promise() for
async/await support

// READ: Fetch all students from MySQL
```

```
app.get('/studentdetails', async (req, res)
=> {
    try {
   const [rows] = await
pool.query('SELECT * FROM
studentdetails');
        res.json(rows);
    } catch (err) {
              res.status(500).json({
                message: err.message
                 });
    }
});

// CREATE: Add a new student
app.post('/studentdetails', async (req, res)
=> {
    const { name,course} = req.body;
    try {
        const [result] = await pool.execute(
            'INSERT INTO studentdetails
(name,course) VALUES (?, ?)',
            [name,course]
```

```
        );
        res.status(201).json({ id:
result.insertId, name,course});
    } catch (err) {
        res.status(500).json({ error:
err.message });
    }
});
// UPDATE: Modify a student record by
ID
app.put('/studentdetails/:id', async (req,
res) => {
    const { id } = req.params;
    const { name,course} = req.body;

    try {
        const [result] = await pool.execute(
    'UPDATE studentdetails SET name =
?,course = ? WHERE id = ?',
            [name,course,id]
        );

        if (result.affectedRows === 0) {
```

```
            return res.status(404).json({
message: "Student not found" });
        }

        res.json({ message: "Student updated
successfully", id, name, course});
    } catch (err) {
        res.status(500).json({ error:
err.message });
    }
});

// DELETE: Remove a student record by
ID
app.delete('/studentdetails/:id', async (req,
res) => {
    const { id } = req.params;

    try {
        const [result] = await
pool.execute('DELETE FROM
studentdetails WHERE id = ?', [id]);
```

```
        if (result.affectedRows === 0) {
            return res.status(404).json({
message: "Student not found" });
        }

        res.json({ message: `Student with ID
${id} deleted successfully` });
    } catch (err) {
        res.status(500).json({ error:
err.message });
    }
});
app.listen(3000, () => console.log('Server
running on port 3000'));
```

Create in MYSQL

1. database : collegedb
2. table : studentdetails with (id, name,course) , id is auto increment and primary key

```
CREATE TABLE studentdetails (
  id INT AUTO_INCREMENT PRIMARY KEY,
```

```
    name VARCHAR(100) NOT NULL,
    course VARCHAR(100) NOT NULL
);
```

Leave space for output

Task 12:

Create a react application for the student management system having registration, login, contact, about pages and implement routing to navigate through these pages.

Solution:

1. Project Setup

First, create your React application and install the routing library.

```
npx create-react-app student-management
cd student-management
npm install react-router-dom
```

2. Main Application & Routing (App.js)

Use the BrowserRouter, Routes, and Route components to define your navigation paths

3. Page Components

Create a pages folder in your src directory , add the following files:

Home.js: start up regarding the student management application .
Register.js: Includes a form for name, email, and password
Login.js: Simple credentials entry.
About.js & **Contact.js**: Static informational pages.

Program :

app.js

```
import React from 'react';

import { BrowserRouter as Router, Routes,
Route, Link } from 'react-router-dom';

import Home from './pages/Home';
```

```
import Register from './pages/Register';
import Login from './pages/Login';
import About from './pages/About';
import Contact from './pages/Contact';

function App() {
  return (
    <Router>
      <nav style={{ padding: '10px',
background: '#333', color: '#fff' }}>
        <Link to="/" style={{ color: '#fff',
margin: '10px' }}>Home</Link>
        <Link to="/register" style={{ color: '#fff',
margin: '10px' }}>Register</Link>
        <Link to="/login" style={{ color: '#fff',
margin: '10px' }}>Login</Link>
        <Link to="/about" style={{ color: '#fff',
margin: '10px' }}>About</Link>
        <Link to="/contact" style={{ color: '#fff',
margin: '10px' }}>Contact</Link>
```

```
      </nav>
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/register"
element={<Register />} />
        <Route path="/login" element={<Login
/>} />
        <Route path="/about" element={<About
/>} />
        <Route path="/contact"
element={<Contact />} />
      </Routes>
    </Router>
  );
}
export default App;
```

Home.js

```
import React from 'react';
```

```
import { Link } from 'react-router-dom';
function Home() {
  const containerStyle = {
    textAlign: 'center',
    padding: '50px',
    fontFamily: 'Arial, sans-serif'
  };
  const buttonStyle = {
    padding: '10px 20px',
    fontSize: '16px',
    backgroundColor: '#007bff',
    color: 'white',
    border: 'none',
    borderRadius: '5px',
    cursor: 'pointer',
    textDecoration: 'none',
    margin: '10px'
  };
```

```
  return (
    <div style={containerStyle}>
      <h1>Welcome to the Student Management
System</h1>
      <p>Manage registrations, logins, and
student profiles efficiently.</p>

</div>
  );
}
export default Home;
```

Register.js

```
export default function Register() {
  return (
    <div>
```

```
      <h2>Student Registration</h2>
      <form>
        <input type="text" placeholder="Full
Name" /><br/>
        <input type="email"
placeholder="Email" /><br/>
        <input type="password"
placeholder="Password" /><br/>
        <button
type="submit">Register</button>
      </form>
    </div>
  );
}
```

Login.js

```
export default function Login() {
  return (
    <div>
```

```
      <h2>Login</h2>
      <input type="email" placeholder="Email"
/><br/>
      <input type="password"
placeholder="Password" /><br/>
      <button>Login</button>
    </div>
  );}
```

About.js

```
export default function About() { return
<h2>About Student Management
System</h2>; }
```

Contact.js

```
export default function Contact() { return
<h2>Contact Support:
admin@school.com</h2>; }
```

Leave space for output